

Exam Date & Time: 25-Nov-2024 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

**Human Interface Interaction [IT-4049]**

**Marks: 50**

**Duration: 180 mins.**

### PART-A

**Answer all the questions.**

- 1)
  - A) Analyze the design of a homepage for a news website that has a cluttered layout with various sections and images competing for attention. Using Gestalt principles, evaluate how the layout could be improved to create a more cohesive and user-friendly interface. Provide specific suggestions based on each principle to enhance visual clarity and usability. (5)
  - B) Write gulf of execution and gulf of evaluation for following the following user goal. "The user wants to mute their microphone during a video conference on Zoom." (3)
  - C) Interpret how monocular cues like size, color, and object placement contribute to depth perception, and explain their role in perceiving three-dimensional space. (2)
- 2)
  - A) Scenario: You are part of a design team tasked with creating a new online ticket booking system for a major sports event. The primary goal is to provide a seamless user experience for purchasing tickets quickly and efficiently. However, there are several constraints to consider, including a tight budget, limited development time, and the need to support a high volume of concurrent users during peak times. Additionally, there are trade-offs between adding advanced features, such as personalized recommendations, and ensuring system reliability under heavy traffic. (5)  
Analyze how you would address the goals, constraints, and trade-offs involved in designing this online ticket booking system. What decisions would you make to balance user experience, performance, and resource limitations? Provide an explanation of how these factors influence the design process.
  - B) Illustrate the importance of the "Golden Rule of Design" in ensuring the effectiveness of design solutions across different contexts. How can understanding both the specific circumstances of a design problem and more generic design principles contribute to creating successful and adaptable designs? (3)
  - C) Interpret how universal design and accessibility considerations contribute to creating inclusive products that cater to diverse user needs. In what ways do assistive technologies enhance the usability of products for individuals with disabilities? (2)
- 3)
  - A) Compare between Parametric and Non-parametric Tests. (5)

- B) Interpret how the different goals of evaluation effect in the design process. How do these goals guide decision-making and improvements in a product or system? (3)
- C) Illustrate any two main responsibilities of the top-level PAC agent. (2)
- 4) Apply the concepts of GOMS (Operators, Methods, and Selection Rules) for the given scenario to analyse which method is preferred by novice and expert users and explain your reasoning.
- A) Scenario: (5)  
A software company offers two ways to save a file:  
1. Method 1: Use the "File" menu → Click "Save"  
2. Method 2: Press Ctrl + S  
The goal is to understand which method suits novice and expert users better.
- B) Explain the role of Backus-Naur Form (BNF) in defining the syntax of programming languages. What are its advantages in the context of language design and interpretation? (3)
- C) Deduce the techniques used in Observational Methods of Evaluation through user participation. (2)
- 5) Apply augmented reality tools in a classroom setting to enhance science education.
- a) Explain how AR can facilitate interaction with educational tools and improve learning outcomes.
- A) b) Assess the benefits of using AR in educational environments for both students and teachers. (5)  
Identify a challenge that may arise when incorporating AR into educational practices.
- B) Interpret the differences between Ethnographic observation vs. experiments. (3)
- C) Illustrate the difficulties faced in evaluating experimental studies on groups. (2)

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